Backend tests: These are test cases for Sprint 1 stories. Stories not part of the Sprint 1 backlog should not be displayed here.

US_01 As the development team, we need clean per-eye labels, a patient-level split and preprocessed photos, so the model can be trained and tested fairly.

TC	TC Title	Step Description	Expected Results	Execution Status	Execution date	Tester
TC_01_01	Verification script passes the real label file	1. Clean clone of main. 2. Run: python scripts/verify_labels.py labels/eye_labels_task7.csv --images "data/Internal/ODIR-5K/Training Images" --manifest data	All checks pass	pass	Oct 4	Anel
TC_01_02	Verification script reports a wrong file as wrong	1. Run: python scripts/verify_labels.py labels/labels_stub.csv	Stub file fails the count and split checks	pass	Oct 4	Anel
TC_01_03	Label file has all columns and 7,000 eyes	1. Run: python scripts/verify_labels.py labels/eye_labels_task7.csv --images "data/Internal/ODIR-5K/Training Images" --manifest data 2. Read the columns and rows lines.	11 columns present; 7,000 rows	pass	Oct 4	Anel
TC_01_04	Per-eye disease counts are correct	1. Same run. 2. Read the count lines.	Cataract 313, glaucoma 282, suspected glaucoma 44, media opacity 63, quality flag 426, excluded 30, normal 2,818, diabetic retinopathy 1,796	pass	Oct 4	Anel
TC_01_05	One-eye and both-condition patient counts	1. Same run. 2. Read the patient count lines.	Cataract in one eye only 111; glaucoma in one eye only 72; both conditions 1	pass	Oct 4	Anel
TC_01_06	Every filename exists and data matches the manifest	1. Same run. 2. Read the filename and MANIFEST lines.	No duplicate filenames; every image found; all 10,324 data files match MANIFEST.sha256	pass	Oct 4	Anel
TC_01_07	No patient in two splits	1. Same run. 2. Read the split lines.	No patient in more than one of train, validation, test	pass	Oct 4	Anel
TC_01_08	Unusable and held-out eyes kept out of training and testing	1. Same run.	No suspected glaucoma, media opacity or excluded eye in train, validation or test	pass	Oct 4	Anel
TC_01_09	Both-condition patient placed in training	1. Same run.	Patient with cataract and glaucoma is in train	pass	Oct 4	Anel
TC_01_10	Split shares are 70 / 15 / 15 by patient	1. Same run.	About 70%, 15%, 15% of patients	pass	Oct 4	Anel
TC_01_11	Every photo has a preprocessed copy	1. Run: python scripts/preprocess_images.py 2. Count files per dataset in Preprocessed/.	ODIR 7,000; ORIGA 650; DRISHTI-GS 101; ACRIMA 705; retina_dataset_2016 601	pass	Oct 4	Anel
TC_01_12	Copies are 512 x 512	1. Read the size of every file in Preprocessed/.	Every copy is 512 x 512 pixels	pass	Oct 4	Anel
TC_01_13	Originals are not changed	1. After preprocessing, re-run the MANIFEST check (verify_labels.py --manifest data).	All 10,324 original files still match their hashes	pass	Oct 4	Anel
TC_01_14	20 random copies look correct by eye	1. Run: python scripts/verify_preprocessing.py 2. Open preprocessing_check.jpg and look at the 20 tiles.	Each tile shows a centred fundus, not cut off or stretched	pass	Oct 4	Anel
TC_01_15	Data pipeline scripts run from a clean clone	1. Fresh clone of main (88d55db). 2. Run every test above.	Scripts run without import errors	pass	Oct 4	Anel

US_02 As a screening technician, I want one photo to give both a cataract and a glaucoma result, so I can triage both conditions in one step.

TC	TC Title	Step Description	Expected Results	Execution Status	Execution date	Tester
TC_02_01	Training smoke test runs end to end	1. Preprocess ODIR (SCRUM-28). 2. Run: python scripts/test_train_dual.py	Training batches and validation batches run with finite loss, no errors	pass	Oct 4	Anel

US_03 As a reviewer, I want the model tested on outside datasets it never trained on, so I can trust that it works beyond ODIR-5K.

TC	TC Title	Step Description	Expected Results	Execution Status	Execution date	Tester
TC_03_01	ORIGA labels	1. Run: python scripts/build_external_labels.py	650 photos, 168 glaucoma	pass	Oct 4	Anel
TC_03_02	DRISHTI-GS labels and spreadsheet cross-check	1. Same run.	101 photos, 70 glaucoma; folder labels agree with the notching spreadsheet	pass	Oct 4	Anel
TC_03_03	ACRIMA labels	1. Same run.	705 photos, 396 glaucoma	pass	Oct 4	Anel
TC_03_04	retina_dataset_2016 labels	1. Same run.	601 photos, 100 cataract, 101 glaucoma	pass	Oct 4	Anel
TC_03_05	Rebuilt label files match the committed ones	1. After the run: git status labels/	No changes to labels/external	pass	Oct 4	Anel
TC_03_06	No identical copy of an ODIR photo in retina_dataset_2016	1. Run: python scripts/check_dataset_overlap.py	Matching fingerprints: 0. The check finds identical files only; resized or edited copies are not detected.	pass	Oct 4	Anel

TC_03_07	A planted identical copy is caught	1. Copy one ODIR photo, renamed, into a test folder next to a retina photo. 2. Point the script at that folder.	Matching fingerprints: 1	pass	Oct 4	Anel
TC_03_08	Calibration tests pass	1. Copy of main (12db1f8). 2. Run: python -m pytest tests/test_calibration_selection.py tests/test_calibration_provenance.py	18 passed	pass	Oct 18	Anel
TC_03_09	Calibration uses the validation split only	1. Copy of main (12db1f8). 2. Read every results/calibration/*_validation.csv and results/calibration/shared/*_validation.csv with its .meta.json. 3. Match the filenames against the validation rows of labels/eye_labels_task7.csv.	9 score files (6 single-disease, 3 shared in shared/), 1,024 rows each; filenames match the 1,024 validation eyes exactly; all marked split = validation; no train, test or external eye	pass	Oct 18	Anel
TC_03_10	External label files are test-only	1. Copy of develop (0f21e1f, PR #43). 2. Read labels/external/*.csv.	2,057 rows across ORIGA, DRISHTI-GS, ACRIMA and retina_dataset_2016; every row has split = test	pass	Oct 16	Anel
TC_03_11	External evaluation protocol is written	1. Copy of develop (0f21e1f, PR #43). 2. Open documents/evaluation/external_validation_protocol.md (SCRUM-44).	Protocol lists the metrics, the freeze rule and the 2,057 test entries; counts match the label files	pass	Oct 16	Anel
TC_03_12	Calibration scripts handle empty results and a worse fit	1. Review and run the Task 12 calibration scripts on develop. 2. Check what happens when no threshold candidate is found and when the Platt fit makes the loss worse.	No empty threshold file is written; a worse fit gives a warning (tiny increase) or an error. Issues found on 11 Oct were fixed in PR #38	pass	Oct 12	Krushi
TC_03_13	Final cutoffs match the approved values	1. Copy of main (12db1f8). 2. Compare results/calibration/frozen_settings.json with the approved CSV (task12_settings_90pct_floor.csv, attached on PR #43).	All 12 model/disease/seed settings (Platt values, cutoff, margin, eyes found, referrals) identical to the approved CSV; the CSV's SHA-256 is recorded in the frozen file	pass	Oct 17	Anel
TC_03_14	Settings are frozen before the low-quality check	1. Open results/calibration/frozen_settings.json. 2. Check the commit order on main: freeze 2920a71, then low-quality report 2688f8c.	status = frozen, frozen = true, dated 17 Oct; review reference PR #43; app model = shared seed 42; frozen before the low-quality check ran	pass	Oct 17	Anel
TC_03_15	Low-quality photo check uses the frozen settings	1. Open results/calibration/final_holdout_report.md and .json. 2. Check its settings SHA-256 against frozen_settings.json. 3. Read the shared seed 42 row.	Runs once on the 21 low-quality eyes (19 patients) with the frozen settings (SHA matches). Shared seed 42: 8 of 21 Uncertain, refer; 15 of 21 referred in total, including those 8. Counts only, no accuracy claim.	pass	Oct 17	Negar (checked by Anel)

US_04 As a screening technician, I want to upload a patient's fundus photo and see a clear cataract and glaucoma result, so I can decide quickly whether the patient should see an eye specialist.

TC	TC Title	Step Description	Expected Results	Execution Status	Execution date	Tester
TC_04_01	Login is required	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Without logging in, the patient list returns 401; a wrong password returns 401	pass	Oct 10	Anel
TC_04_02	Logout ends the session	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	After logout, the current-user call returns 401	pass	Oct 10	Anel
TC_04_03	Demo data loads	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	4 made-up patients sorted by last name; 5 screenings listed newest first	pass	Oct 10	Anel
TC_04_04	New patient gets the next ID	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Next ID is P-005; a patient with an empty name is rejected (422)	pass	Oct 10	Anel
TC_04_05	A fundus photo is screened	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Photo passes the quality check; screening returns a cataract and a glaucoma result, an overall result and both heatmaps; model is marked as mock	pass	Oct 10	Anel
TC_04_06	A failed photo is never screened	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	A screenshot fails as not_fundus and cannot be screened (409); a non-image file fails as unreadable	pass	Oct 10	Anel
TC_04_07	Photo must match the patient and eye	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Screening a right-eye photo as the left eye is refused (409)	pass	Oct 10	Anel
TC_04_08	The same eye can be screened again	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Follow-up screenings are all kept and the newest one counts as latest	pass	Oct 10	Anel (added after Krushi's review)
TC_04_09	The same photo is not screened twice	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	A second upload of the same photo is refused and points to the first screening	pass	Oct 10	Anel (added after Krushi's review)
TC_04_10	Database allows one screening per photo	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	A second screening of the same photo is blocked by the database itself	pass	Oct 10	Anel (added after Krushi's review)
TC_04_11	Skipped eye and today's visit	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Skipping an eye needs a reason; the skipped eye shows in today's visit with no result	pass	Oct 10	Anel
TC_04_12	History search	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Searching 'lopez' finds only that patient's screenings; filtering by patient works	pass	Oct 10	Anel
TC_04_13	Exports are PDFs	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Screening and patient exports download as PDF files	pass	Oct 10	Anel

TC_04_14	Unknown IDs return 404	1. On branch task-api-mock (PR #35). 2. Run: python -m pytest tests/test_api.py -v	Unknown patient, screening, heatmap and image IDs return 404	pass	Oct 10	Anel

US_05 As a screening technician, I want the app to stop photos that can't be screened and tell me why, so I don't get a result from an unusable file.

TC	TC Title	Step Description	Expected Results	Execution Status	Execution date	Tester
TC_05_01	Fundus-like photo passes	1. Run: pytest tests/test_quality_check.py -v	passed	pass	Oct 4	Anel
TC_05_02	Fundus circle cut off top and bottom passes (ORIGA style)	1. Run: pytest tests/test_quality_check.py -v	passed	pass	Oct 4	Anel
TC_05_03	Photo with white border passes	1. Run: pytest tests/test_quality_check.py -v	passed	pass	Oct 4	Anel
TC_05_04	Dim, hazy photo passes (cataract look)	1. Run: pytest tests/test_quality_check.py -v	passed	pass	Oct 4	Anel
TC_05_05	Blurry photo passes	1. Run: pytest tests/test_quality_check.py -v	passed	pass	Oct 4	Anel
TC_05_06	Real ODIR photos pass (incl. washed-out cataract, very dark glaucoma)	1. Run: pytest tests/test_quality_check.py -v	passed	pass	Oct 4	Anel
TC_05_07	Unreadable bytes fail	1. Run: pytest tests/test_quality_check.py -v	failed: unreadable	pass	Oct 4	Anel
TC_05_08	Empty file fails	1. Run: pytest tests/test_quality_check.py -v	failed: unreadable	pass	Oct 4	Anel
TC_05_09	Truncated JPEG fails	1. Run: pytest tests/test_quality_check.py -v	failed: unreadable	pass	Oct 4	Anel
TC_05_10	Almost black photo fails	1. Run: pytest tests/test_quality_check.py -v	failed: too dark	pass	Oct 4	Anel
TC_05_11	Blank white photo fails	1. Run: pytest tests/test_quality_check.py -v	failed: not fundus	pass	Oct 4	Anel
TC_05_12	Blank grey photo fails	1. Run: pytest tests/test_quality_check.py -v	failed: not fundus	pass	Oct 4	Anel
TC_05_13	Text document fails	1. Run: pytest tests/test_quality_check.py -v	failed: not fundus	pass	Oct 4	Anel
TC_05_14	Random noise fails	1. Run: pytest tests/test_quality_check.py -v	failed: not fundus	pass	Oct 4	Anel
TC_05_15	Blue photo fails	1. Run: pytest tests/test_quality_check.py -v	failed: not fundus	pass	Oct 4	Anel
TC_05_16	Tight crop without camera border fails	1. Run: pytest tests/test_quality_check.py -v	failed: not fundus	pass	Oct 4	Anel
TC_05_17	Every failure reason has a message for the user	1. Run: pytest tests/test_quality_check.py -v	unreadable, too dark, not fundus all have text	pass	Oct 4	Anel

US_06 As the development team, we want cataract-only, glaucoma-only and shared two-head models trained with identical settings, so we can test whether sharing the backbone helps.

TC	TC Title	Step Description	Expected Results	Execution Status	Execution date	Tester
TC_06_01	Glaucoma-only training smoke test	1. Copy of main (efe7789) with Preprocessed/ODIR. 2. Run: python scripts/test_train_glaucoma.py	Glaucoma Task 11 smoke test passed	pass	Oct 16	Anel
TC_06_02	Cataract-only training smoke test	1. Copy of main (efe7789) with Preprocessed/ODIR. 2. Run: python scripts/test_train_cataract.py	Cataract Task 11 smoke test passed	pass	Oct 16	Anel
TC_06_03	Shared two-head training pipeline	1. Copy of main (efe7789) with Preprocessed/ODIR. 2. Run test_training_pipeline() from scripts/test_train_dual_task11.py (synthetic data; the full script also needs the 3 shared checkpoints downloaded).	Two-head training, evaluation, mean AUROC, reload and results update all pass	pass	Oct 16	Anel
TC_06_04	Nine models are saved	1. Open the GitHub releases task11-cataract-checkpoints, task11-glaucoma-checkpoints and task11-shared-checkpoints.	9 checkpoint files: 3 model types x seeds 42, 43, 44	pass	Oct 16	Anel
TC_06_05	Checkpoints match the recorded hashes	1. Compare the SHA-256 column of results/task11/README.md with the release file digests.	9 of 9 match	pass	Oct 16	Anel
TC_06_06	Best epoch matches the training history	1. For each model type, compare results/task11/*_results.csv with *_training_history.csv.	Best epoch and AUROC match for all 9 runs (glaucoma history is rounded to 4 decimals); each run stops 5 epochs after its best epoch	pass	Oct 16	Anel
TC_06_07	Same settings for all nine runs	1. Read learning_rate, batch_size, max_epochs and patience in the three *_results.csv files.	Learning rate 0.0001, batch 8, max 30 epochs, patience 5 for all 9 runs	pass	Oct 16	Anel
TC_06_08	Only the training split is used to learn; no external data	1. Read scripts/fundus_dataset.py, train_glaucoma.py, train_cataract.py and train_dual_task11.py.	Training reads only the train split, validation decides when to stop, images come from Preprocessed/ODIR; no external dataset in any training script	pass	Oct 16	Anel

DUAL Sprint 1 test cases: frontend

US_04 As a screening technician, I want to upload a patient's fundus photo and see a clear cataract and glaucoma result, so I can decide quickly whether the patient should see an eye specialist.

TC	TC Title	Step Description	Expected Results	Execution Status	Tester
TC_04_15	Select patient and eye	Open Upload. Choose a demo patient and select Left or Right.	Selected patient and eye appear before screening.	pass	Bhoami
TC_04_16	Open main pages	Use navigation to open Home, Upload, Results, History and About.	Available pages open with the correct content.	pass	Bhoami
TC_04_17	Upload and preview a photo	Choose a JPG/PNG fundus photo. Review its preview before screening.	Selected photo appears in the upload preview.	pass	Bhoami
TC_04_18	Reject unsupported files	Try uploading a non-image file, such as PDF or TXT.	Unsupported file is rejected with clear feedback.	pass	Bhoami
TC_04_19	Reject poor-quality image	Upload a poor-quality fundus photograph and submit for screening.	Quality warning appears; normal screening does not proceed.	pass	Bhoami
TC_04_20	Show screening results	Submit an accepted fundus photo and open the Results page.	Overall result and separate cataract/glaucoma results appear.	pass	Bhoami
TC_04_21	Display risk information	Review the condition scores, thresholds, and result details.	Risk information is shown for available mock-model outputs.	pass	Bhoami
TC_04_22	Handle Refer / Uncertain	Review screens for Refer and Uncertain outputs when returned.	Correct label and follow-up guidance are displayed.	pass	Bhoami
TC_04_23	Review screening history	Open History after completing a screening.	Stored screening records can be retrieved and reviewed.	pass	Bhoami
TC_04_24	Switch light/dark theme	Use the theme toggle on the application interface.	Theme changes and content remains readable.	pass	Bhoami
TC_04_25	Check mobile layout	Open Home, Upload, Results and History at phone width.	Content fits without cutoff or unwanted horizontal scrolling.	pass	Bhoami
TC_04_26	Use demo patient records	Review the patient selection and displayed patient details.	Prototype uses fictional patient records, not real patient data.	pass	Bhoami

DUAL Sprint 1 test summary

Story ID	Feature	Area	Total	Passed	Failed	Not run
US_01	Data Pipeline	Backend	15	15	0	0
US_02	Dual-Head Model	Backend	1	1	0	0
US_03	External Test Sets	Backend	15	15	0	0
US_04	App Prototype (API)	Backend	14	14	0	0
US_04	App Prototype (screens)	Frontend	12	12	0	0
US_05	Photo Quality Check	Backend	17	17	0	0
US_06	Model Ablation	Backend	8	8	0	0
Total			**82**	**82**	**0**	**0**

Sprint 1 test cases. Totals update automatically from the Backend tests and Frontend tests tabs.